

平均场动力学

从粒子到测度， 再回到粒子

平均场动力学的学习与数据同化

Liyao Lyu

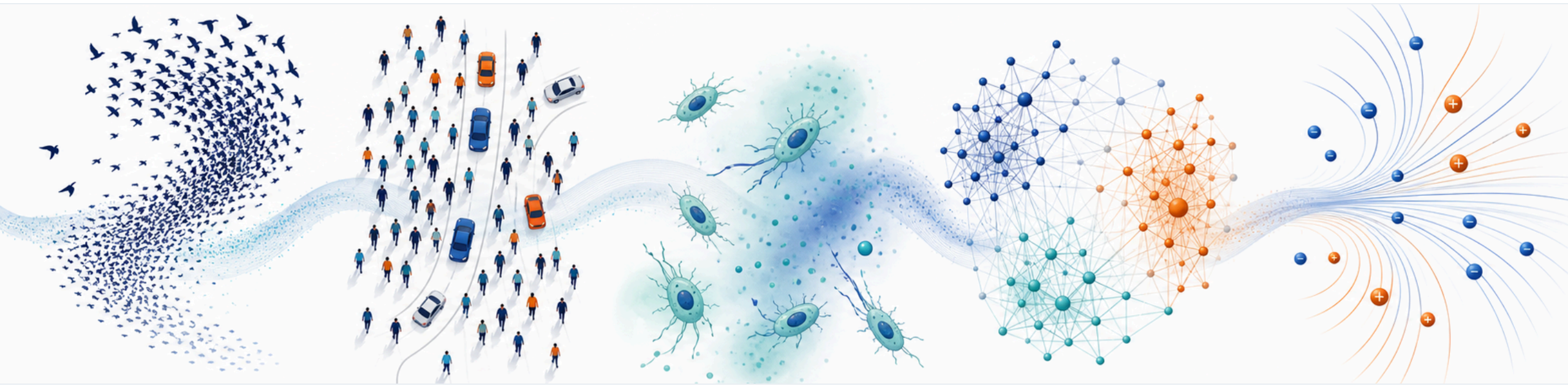
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Mean-field models connect systems across science

When many similar agents interact through the population, their collective behavior is naturally described by an evolving distribution.



Collective motion

bird flocks · fish schools · swarms

Crowds & transport

congestion · lane formation

Living systems

cells · chemotaxis · neural populations

Social dynamics

consensus · clustering · polarization

Kinetic physics

plasmas · particle beams

Different mechanisms, same reduction: many labeled agents $\longrightarrow$ one evolving law μ_t .

From Particles to Waves and Back · Lecture 11, HST6

Mean-field reduction: discard labels, then let $N \rightarrow \infty$

1 · LABELED SYSTEM

N interacting agents

$$d\mathbf{X}_t^i = \frac{1}{N} \sum_{j=1}^N K(\mathbf{X}_t^j - \mathbf{X}_t^i) dt + \Sigma d\mathbf{W}_t^i.$$

Discard the agent labels

2 · MEASURE-VALUED STATE

Keep only the empirical measure

$$\mu_t^N = \frac{1}{N} \sum_{j=1}^N \delta_{\mathbf{X}_t^j}$$

The interaction depends on the population only through μ_t^N .
Relabeling or reordering the agents leaves the dynamics unchanged.

Let $N \rightarrow \infty$

3 · MEAN-FIELD LIMIT

One particle coupled to its

law From Particles to Measures and Back · Liyao Lyu, USTC

$$d\mathbf{X}_t = \mathbf{b}(\mathbf{X}_t, \mu_t) dt + \Sigma d\mathbf{W}_t, \quad \mu_t = \text{Law}(\mathbf{X}_t).$$

A three-step loop from particle data to corrected forecasts

PART I

Learn

Unknown dynamics

Particle trajectories $\longrightarrow \mathbf{b}_\theta(\mathbf{x}, \mu)$ 

MODEL ROLLOUT

Forecast

Evolve the population

Mean-field model $\longrightarrow$
 N -particle forecast

PARTS II-III

Correct

Forecast drift

Macroscopic data $\longrightarrow$
particle nudging

Learning the drift from particle data

Learn $\mathbf{b}(\mathbf{x}, \mu)$ directly from trajectories as a function of the population law.

Why learning $\mathbf{b}(\mathbf{x}, \mu)$ is hard

GIVEN BY PARTICLE TRAJECTORIES

Particle configurations evolving in time

The data tell us where the particles move.



UNKNOWN · LEARN FROM DATA

$$\mathbf{b} : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}^d$$

How the population law changes a particle's drift.

THE MODEL CLASS MUST OVERCOME THREE OBSTACLES

01 · COMPUTATION

Direct interactions are expensive

$$\mathbf{b}_i = \frac{1}{N} \sum_{j=1}^N K(\mathbf{X}_i, \mathbf{X}_j), \quad i = 1, \dots, N$$

all drifts: $O(N^2)$

Evaluating every drift touches every particle pair.

02 · EXPRESSIVITY

Pairwise models are linear in the measure

$$\mathbf{b}(\mathbf{x}, \mu) = \int K(\mathbf{x}, \mathbf{y}) \mu(d\mathbf{y})$$

linear in μ

They cannot express nonlinear many-body effects beyond additive pairs.

03 · INPUT SPACE

The population input is a probability measure

Standard MLP

$$F_\theta : \mathbb{R}^{Nd} \rightarrow \mathbb{R}^{Nd}, \quad \mathbf{X} \mapsto \mathbf{b}$$

fixed dimension Nd · order-sensitive

Operator learning

$$\mathcal{G}_\theta : \mathcal{A} \rightarrow \mathcal{U}, \quad a(\cdot) \mapsto u(\cdot)$$

function space $\longrightarrow$ function space

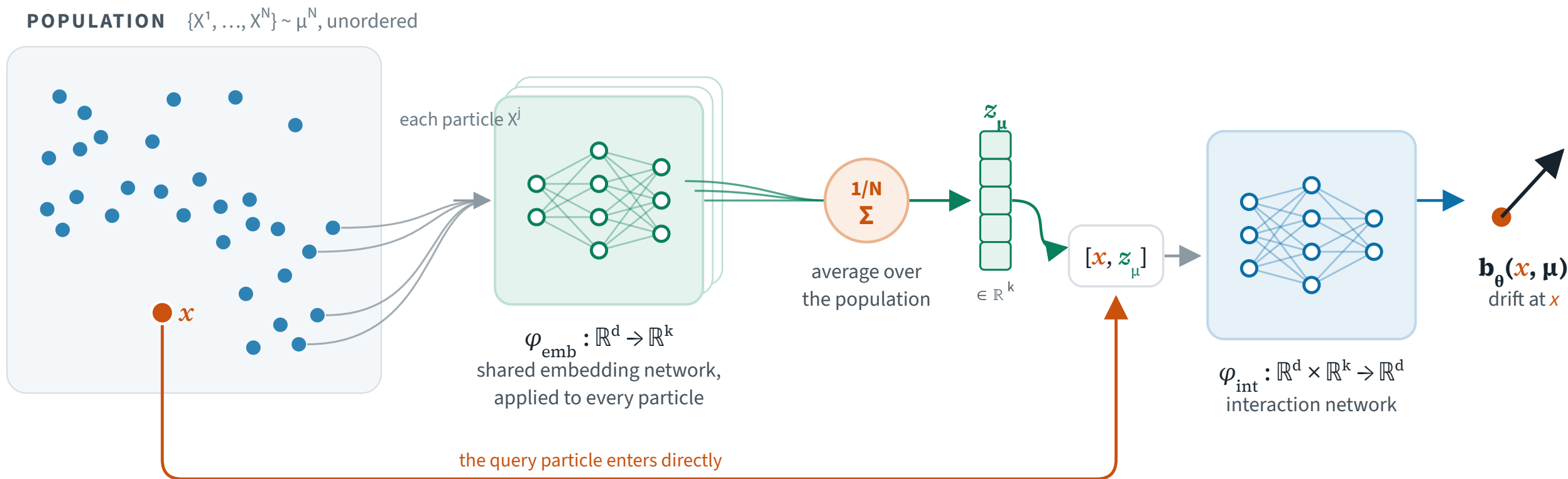
Mean-field drift

$$\mathbf{b} : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}^d$$

state + probability measure $\longrightarrow$ drift

Measure-Valued Neural Network (MVNN)

Two neural networks connected by one population average: $\mathbf{b}_\theta(\mathbf{x}, \mu) = \varphi_{\text{int}}(\mathbf{x}, \langle \varphi_{\text{emb}}, \mu \rangle)$.



Key idea. The population enters only through the averaged feature $\mathbf{z}_\mu$ — permutation invariant by construction, one embedding pass per particle: $O(N)$, not $O(N^2)$.

The learned drift defines a genuine mean-field model

PROPOSITION Well-posedness of the learned dynamics

If φ_{emb} and φ_{int} are globally Lipschitz and $f_0 \in \mathcal{P}_2(\mathbb{R}^d)$, then for any $T > 0$ the learned McKean–Vlasov SDE

$$d\mathbf{X}_t^\theta = \mathbf{b}_\theta(\mathbf{X}_t^\theta, f_t^\theta) dt + \sigma d\mathbf{B}_t, \quad f_t^\theta = \text{Law}(\mathbf{X}_t^\theta),$$

has a unique strong solution on $[0, T]$, and f_t^θ is the unique weak solution of the associated nonlinear Fokker–Planck equation. **Meaning:** trained weights give a well-defined continuum model — no smoothing or filtering of the data is needed.

PROPOSITION Propagation of chaos for the learned particle system

Under the same assumptions, let $(\mathbf{X}_t^{\theta, i, N})_{i=1}^N$ solve the learned N -particle system with f_0 -chaotic initial data. Then for any $T > 0$ the particle law is f^θ -chaotic:

$$\lim_{N \rightarrow \infty} W_2(f_{[0, T]}^{1, \theta, N}, f_{[0, T]}^\theta) = 0.$$

Meaning: the N -particle simulation and the mean-field model agree as N grows — train at one N , simulate at any N .

How expressive is the architecture?

THEOREM Universal approximation (after Pham–Warin)

Let ζ be a probability measure on $\mathcal{P}_2(\mathbb{R}^d)$ and $\mathbf{b}^* : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}^d$ continuous with $\|\mathbf{b}^*\|_{L^2(\zeta)} < \infty$. For every $\varepsilon > 0$ there exist $k \in \mathbb{N}$ and networks $\varphi_{\text{emb}} : \mathbb{R}^d \rightarrow \mathbb{R}^k$, $\varphi_{\text{int}} : \mathbb{R}^d \times \mathbb{R}^k \rightarrow \mathbb{R}^d$ such that

$$\|\mathbf{b}^* - \varphi_{\text{int}}(\cdot, \langle \varphi_{\text{emb}}, \cdot \rangle)\|_{L^2(\zeta)} \leq \varepsilon.$$

THEOREM Quantitative approximation via finite-dimensional measure embeddings

Let $f : \Omega \times U \rightarrow \mathbb{R}$ be Lipschitz in the state and in W_1 , with all measures supported in $\Omega = [-\gamma_1, \gamma_1]^d$. Cover Ω by $C_\Omega = \varepsilon^{-d}$ balls, take a partition of unity $\{\omega_m\}$ subordinate to it, and embed $\mu \mapsto \mathbf{u} = (\langle \omega_1, \mu \rangle, \dots, \langle \omega_{C_\Omega}, \mu \rangle)$. Then there are coefficients a_k and ReLU networks q_k with explicit depth, sparsity and weight bounds such that

$$\sup_{\mathbf{x} \in \Omega, \mu \in U} \left| f(\mathbf{x}, \mu) - \sum_{k=1}^H a_k q_k(\mathbf{x}, \mathbf{u}) \right| \leq \varepsilon, \quad H = O\left(((C_\Omega + d) C_\Omega d)^{\frac{C_\Omega + d}{2}} \varepsilon^{-C_\Omega - d} \right).$$

Meaning: exactly MVNN's form — a finite embedding of μ feeding an interaction network — made fully constructive. But a *fixed* histogram-type embedding needs $C_\Omega = \varepsilon^{-d}$ features, and H blows up accordingly: the curse of dimensionality when $\mathbf{b}$ may be an arbitrary functional of μ .

Approximation rate under low-dimensional measure dependence

ASSUMPTION Finite-dimensional measure dependency

On the family U of measures of interest, every functional is determined by r fixed feature functions:

$$V(\mu) = G(\langle g_1, \mu \rangle, \dots, \langle g_r, \mu \rangle) \quad \forall \mu \in U.$$

Meaning: U lies on a finite-dimensional manifold inside $\mathcal{P}_2(\mathbb{R}^d)$. Gaussians are the model case — determined by mean and covariance, $r = d + d(d + 1)/2$; flocking, consensus and aggregation are governed by a few order parameters.

THEOREM Approximation rate of MVNN under the assumption

Suppose the drift depends on the measure only through the r features, $\mathbf{b}^*(\mathbf{x}, \mu) = G(\mathbf{x}, \langle g_1, \mu \rangle, \dots, \langle g_r, \mu \rangle)$ on $[0, 1]^d$ with G and the g_i Lipschitz. Then for every $\varepsilon > 0$ there are deep ReLU networks achieving $\sup_{\mathbf{x}, \mu} \|\mathbf{b}^* - \mathbf{b}_\theta\| \leq \varepsilon$ with

$$\text{depth} = O(\log \varepsilon^{-1}), \quad W_{\text{int}} = O(\varepsilon^{-(d+r)}), \quad W_{\text{emb}} = O(\varepsilon^{-d}).$$

Meaning: the embedding network *learns* the r order parameters instead of the ε^{-d} fixed features of the previous theorem — the complexity is governed by $d + r$, and the curse disappears.

From Particles to Measures and Back, Liang and NDTG

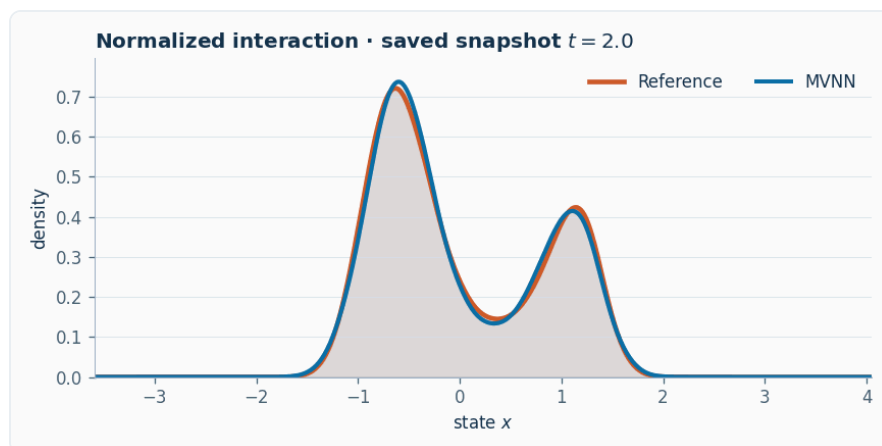
MVNN captures normalized interactions

$$\dot{X}_t^i = \frac{\sum_j \phi(|X_t^j - X_t^i|) (X_t^j - X_t^i)}{\sum_k \phi(|X_t^k - X_t^i|)}, \quad \phi(r) = e^{-(r/\ell)^2}, \quad \ell = 0.5.$$

Design. The normalization makes **b nonlinear in μ** , with asymmetric weights.

Train. $N = 16,000, M = 100, T = 2, \Delta t = 10^{-2}$; random Gaussian mixtures (2–8 modes).

Test. Autonomous rollout of the full N -particle system from **unseen** mixtures.



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MVNN tracks the reference from an unseen initial law.

- **WSINDy**: additive $\int K(y - x) \mu(dy)$ — linear in μ .
- **Online param. estimation**: needs the true form of **b** + a good initial guess.

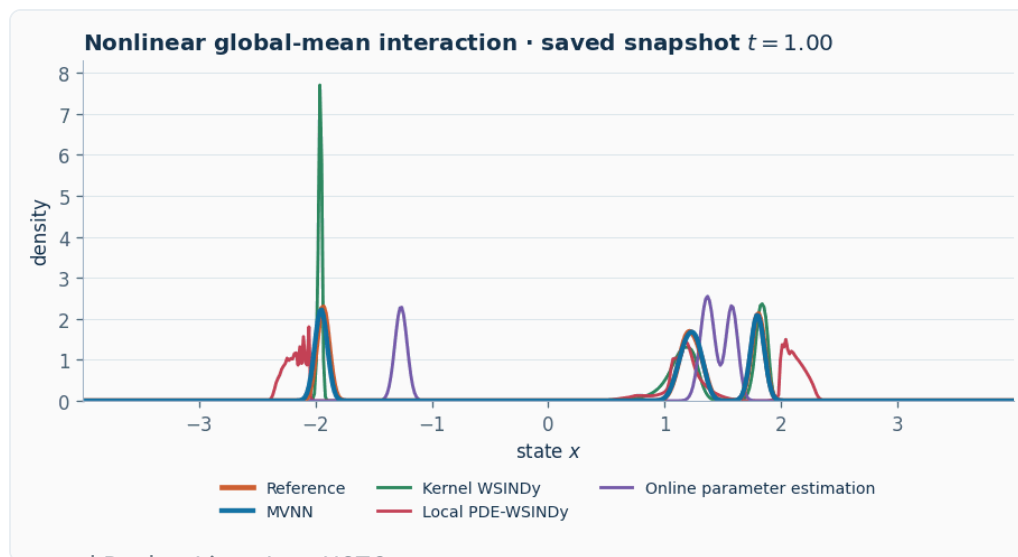
MVNN finds a global order parameter

$$dX_t^i = [\gamma(M_t^N - X_t^i) + (a - \alpha(M_t^N)^2)X_t^i - \beta(X_t^i)^3]dt + \sigma dW_t^i, \quad M_t^N = \frac{1}{N} \sum_j X_t^j.$$

Design. The drift couples to the population through M_μ and $(M_\mu)^2$ — **quadratic in the measure**.

Train. $a = \alpha = 1, \beta = \gamma = 0.2, \sigma = 0.03, N = 1024$; random mollifier-mixture initial laws.

Test. 36 held-out trajectories; band = three independently trained MVNNs.



METHOD	AVG. $\bar{W}_1$
MVNN	0.005–0.006
kernel WSINDy	0.091
local PDE-WSINDy	0.27
online param. est.	0.18

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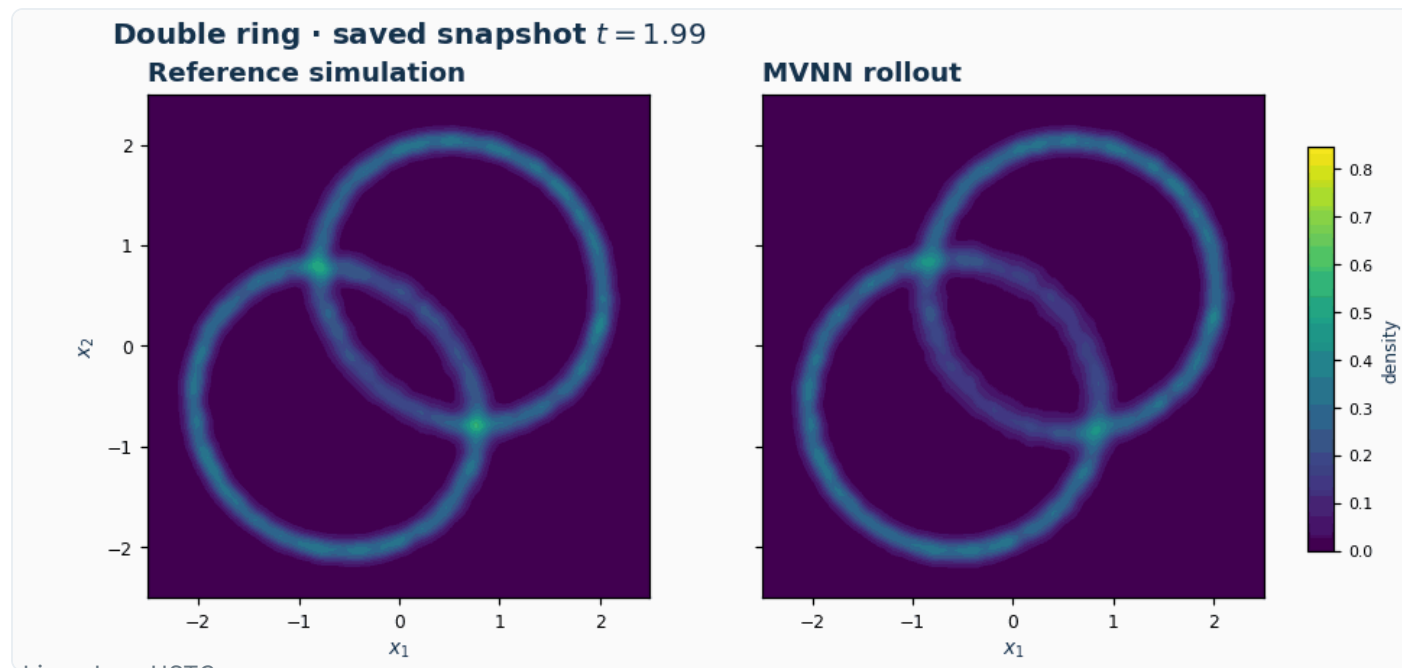
Only MVNN follows the nonlinear global-mean evolution.

Generalization in 2D: unseen initial geometries

Design. Attraction–repulsion swarm: $\dot{\mathbf{X}}^i = \frac{1}{N} \sum_j \phi(\|\mathbf{X}^j - \mathbf{X}^i\|)(\mathbf{X}^j - \mathbf{X}^i)$, $\phi(r) = c_{\text{rep}} e^{-(r/\ell_{\text{rep}})^2} - c_{\text{att}} e^{-(r/\ell_{\text{att}})^2}$; $N = 16,000$.

Train. Only noisy single rings, of random radius and width.

Test. Double-ring, uniform-disk and binary asymmetric μ_0 — same law, only the initial geometry changes.



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Double ring — unseen geometry; contraction and topology stay synchronized.

Extend to a multi-species setup

Asymmetric three-species model

$$\dot{X}_t^{i,k} = \sum_{\ell=1}^3 \frac{1}{N_\ell - \delta_{k\ell}} \sum_{\substack{j \\ (j,\ell) \neq (i,k)}} \phi_{k\ell}(|X_t^{i,k} - X_t^{j,\ell}|) (X_t^{j,\ell} - X_t^{i,k}),$$

$$\phi_{k\ell}(r) = D_{k\ell} \psi(r/R_\ell), \quad \psi(u) = \exp\left(1 - \frac{1}{1 - |u|^{10}}\right) \mathbf{1}_{\{|u| < 1\}}.$$

$$D = \begin{pmatrix} 5 & 10 & 0 \\ 0 & 2 & 5 \\ 0 & 0 & 1 \end{pmatrix}, \quad (R_1, R_2, R_3) = (1, 2.5, 5).$$

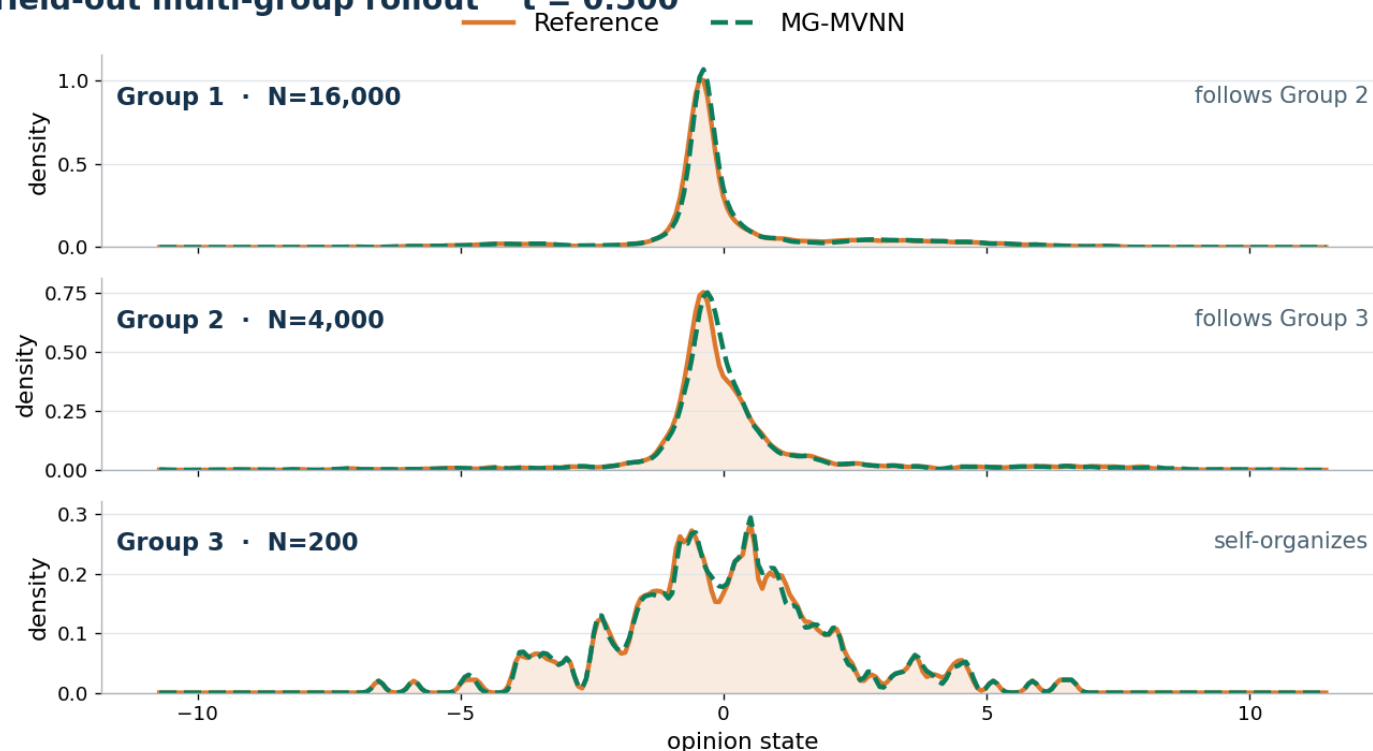
$D_{k\ell}$: source species ℓ acts on target species k ; R_ℓ is its interaction radius. Thus **3** $\rightarrow$ **2** $\rightarrow$ **1**.

MG-MVNN. Learn $b_{\theta,k}(x, \mu_1, \mu_2, \mu_3)$ for each species; architecture details are in the paper.

Held out. Spatially separated species with

$(N_1, N_2, N_3) = (16,000, 4,000, 200)$.

Held-out multi-group rollout $t = 0.500$



*MG-MVNN reproduces the hierarchical entrainment **3** $\rightarrow$ **2** $\rightarrow$ **1**.*

Phase-space MVNN learns acceleration

MVNN learns the acceleration

$$\dot{X}_t^i = V_t^i, \quad \dot{V}_t^i = b_\theta(X_t^i, V_t^i, \mu_t^N), \quad \mu_t^N = \frac{1}{N} \sum_j \delta_{(X_t^j, V_t^j)}.$$

$$b_\theta(x, v, \mu) = \varphi_{\text{int}}\left(x, v, \int \varphi_{\text{emb}}(y, w) \mu(dy, dw)\right).$$

Target: $A_\ell^i \approx (V_{\ell+1}^i - V_\ell^i)/\Delta t$.

Cucker-Smale test system

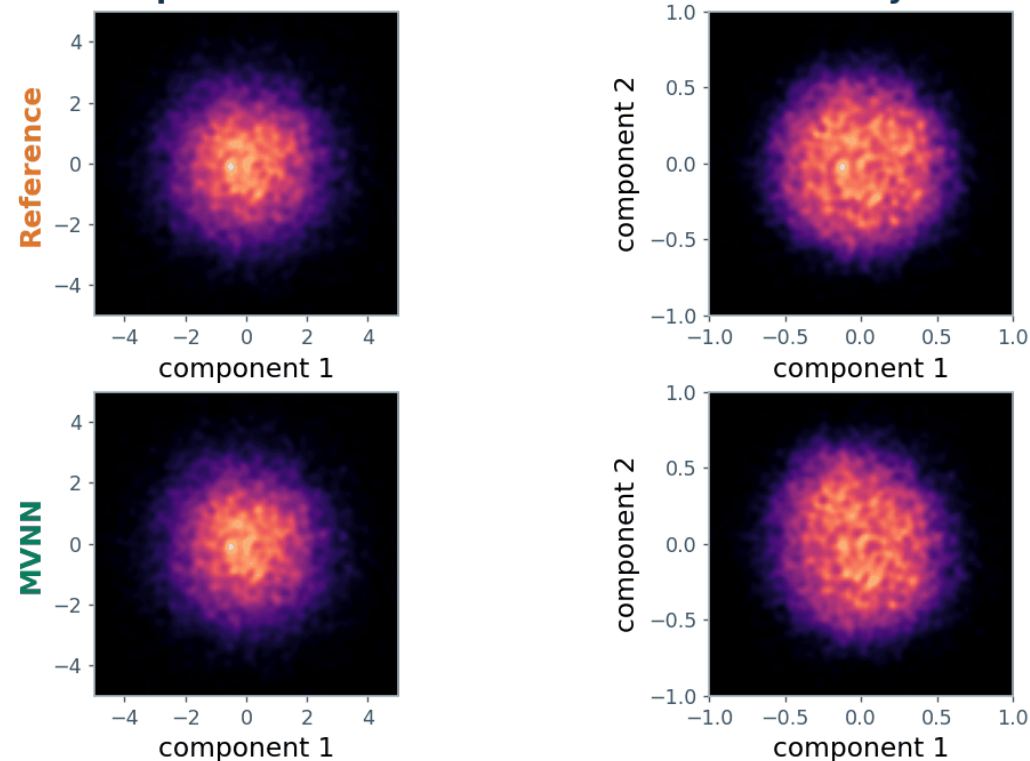
$$\dot{X}_t^i = V_t^i,$$

$$\dot{V}_t^i = \frac{1}{N} \sum_{j=1}^N \phi(\|X_t^j - X_t^i\|) (V_t^j - V_t^i),$$

$$\phi(r) = c_{\text{rep}} e^{-(r/\ell_{\text{rep}})^2} - c_{\text{att}} e^{-(r/\ell_{\text{att}})^2}.$$

$(c_{\text{rep}}, \ell_{\text{rep}}, c_{\text{att}}, \ell_{\text{att}}) = (1, 0.5, 0.7, 2)$, $N = 16,000$

One held-out Cucker-Smale rollout **N = 16,000** **t = 2.0**
position **x** velocity **v**

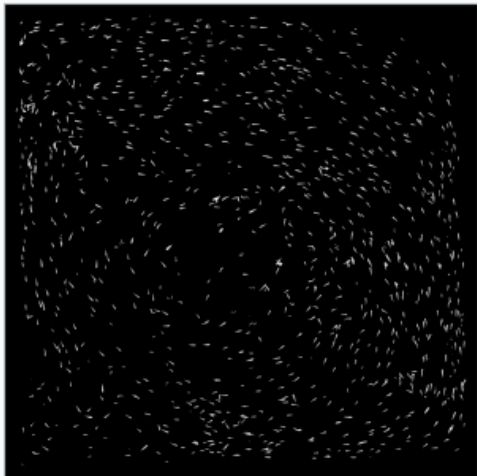


On held-out data, MVNN reproduces both position and velocity distributions.

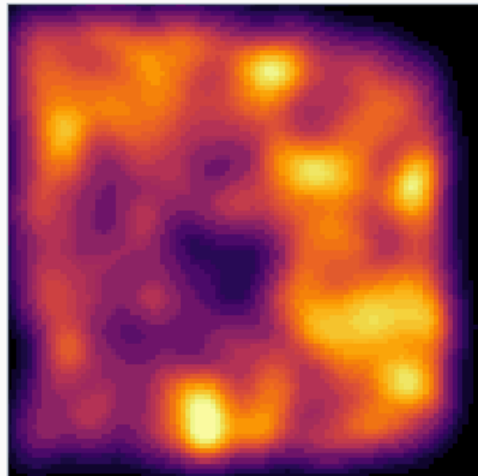
A learned model still drifts

Fish school · saved frame $t = 6.25$ s

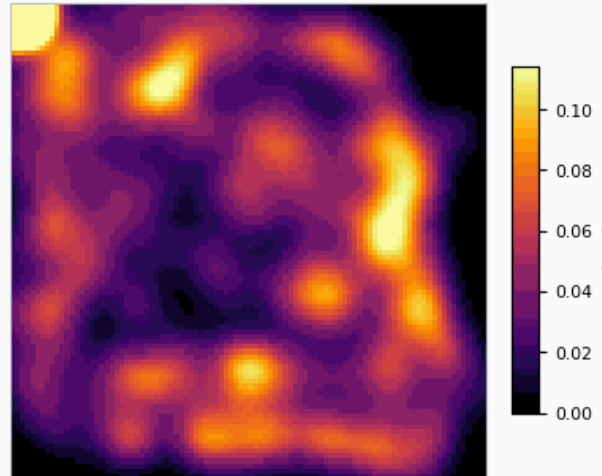
Experiment video



Observed density



Learned free simulation



The learned model initially captures the circulation, then drifts away from the experiment.

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Observation: a measure

$N \approx 1,126$ fish; occlusions and identity dropouts make the usable observation an unlabeled KDE density.

Simulation: open loop

The learned drift captures circulation initially, but bias accumulates and the free simulation **loses the observed school within seconds.**

Needed. Compare distributions, then nudge particles.

Multiscale Nudging

Compare forecasts and data as measures; correct the particles directly.



Density data have no particle labels

GIVEN

A biased forecast + coarse density observations

Forecast: labeled particles $\mathbf{Y}_t^i$. Data: the smoothed law $K_h * \mu_t$ — unlabeled, macroscopic.



WANTED

A correction applied to every particle

Driven by the measure-level mismatch — no labels, no matching.

ONE MISMATCH BEHIND ALL OF THEM: THE DATA CARRY NO PARTICLE LABELS

μ^{obs} is a **permutation-invariant functional of the empirical distribution** — it says where the mass is, never which particle is which. Each classical update needs those labels anyway:

- **EnKF** — the update is built from covariances between *labeled* state coordinates, so it depends on particle indexing unless identities, an ordering, or a matching rule are imposed.
- **3D/4D-Var** — handles nonlinear observation operators, but fitting a coarse density means optimizing over the configurations that produce it: the lift from density back to labeled particles is highly non-unique.
- **Particle filters** — the importance weights are likelihoods of *labeled* states; weight degeneracy in high dimension on top.
- **Transport / mean-field filters** — feedback velocities built from a likelihood on the labeled state, not on the density.

Nudging is the exception — a simple feedback with rigorous theory, but classically formulated on labeled states and fields. **Part II lifts it to the measure level.**

Truth, model, and observations

True process $\mathbf{X}_t$ and model $\mathbf{Y}_t$: shared noise channel, bias in the drift

$$d\mathbf{X}_t = \mathbf{b}_{\text{true}}(\mathbf{X}_t, \mu_t)dt + \Sigma d\mathbf{W}_t, \quad d\mathbf{Y}_t = \mathbf{b}_{\text{model}}(\mathbf{Y}_t, \nu_t)dt + \Sigma d\mathbf{W}_t, \quad \mathbf{R} := \mathbf{b}_{\text{model}} - \mathbf{b}_{\text{true}}.$$

Smoothing kernel $K_h(\mathbf{x}) = h^{-d}K(\mathbf{x}/h)$ (Gaussian), even; the bandwidth h is the resolution of the measurement device.

Case A — pixel / density observation

$$\mu_t^{\text{obs}}(\mathbf{z}) = (K_h * \mu_t)(\mathbf{z}).$$

Imaging, satellite products, reconstructed densities.

Case B — unlabeled particle locations

$$\mu_t^{\text{obs}}(\mathbf{z}) = \frac{1}{M} \sum_{k=1}^M K_h(\mathbf{z} - \mathbf{x}_t^{\text{obs},k,M}).$$

Tracking with dropouts, no index matching across frames.

Both cases enter the same framework through the same object, the **measure** μ^{obs} — and the forecast is compared with it only after the same smoothing K_h .

Nudging is a Wasserstein gradient

Misfit at the observation scale, and its first variation (K_h even $\Rightarrow$ self-adjoint)

$$J_{\text{nud},h}(\nu, \mu^{\text{obs}}) = \frac{1}{2} \int_{\mathbb{R}^d} |K_h * \nu - \mu^{\text{obs}}|^2 d\mathbf{x}, \quad \phi_{\nu,h} := \frac{\delta J_{\text{nud},h}}{\delta \nu} = \widetilde{K}_h * \nu - K_h * \mu^{\text{obs}}, \quad \widetilde{K}_h := K_h * K_h.$$

In Wasserstein geometry, the first variation gives the steepest-descent direction:

$$\mathbf{u}_{\text{nud}}(\mathbf{z}) = -\lambda \nabla \phi_{\nu,h}(\mathbf{z}) = -\lambda \nabla (\widetilde{K}_h * \nu - K_h * \mu^{\text{obs}})(\mathbf{z}).$$

The same field drives the finite particles, the representative mean-field particle, and its law.

Compare at the density scale, correct at the particle scale

MACROSCOPIC — OBSERVATION SCALE h

observed data (true system): image / pixels, or
unlabeled locations $\rightarrow$ KDE $\rightarrow \mu^{\text{obs}}$

coarse-grained forecast

$$K_h * \nu = \frac{1}{N} \sum_j K_h(\cdot - \mathbf{Z}^j)$$

compare at scale h

$$\text{residual } \mathbf{r} = K_h * \nu - \mu^{\text{obs}}, \text{ misfit } J = \frac{1}{2} \int |\mathbf{r}|^2$$



coarse-grain — permutation invariant, cost $O(NG)$

Wasserstein-gradient nudge $\mathbf{U}^i = -\lambda \nabla (\widetilde{K}_h * \nu - K_h * \mu^{\text{obs}})(\mathbf{Z}^i)$



MICROSCOPIC — PARTICLE SCALE

forecast particles $\{\mathbf{Z}^i\}_{i=1}^N \sim \nu$

forecast step (biased model) $\mathbf{Z} \leftarrow \mathbf{Z} + \Delta t \mathbf{b}_{\text{model}}(\mathbf{Z}, \nu) + \Sigma \sqrt{\Delta t} \xi$, then L_{nud} nudging
substeps $\hookrightarrow$ repeat

No particle matching, no model linearization, and no ensemble covariances.

Mean-field particle dynamics are well posed

Mean-field particle dynamics

$$d\mathbf{Z}_t = \mathbf{b}_{\text{model}}(\mathbf{Z}_t, \nu_t)dt - \lambda \nabla (\widetilde{K}_h * \nu_t - K_h * \mu_t^{\text{obs}})(\mathbf{Z}_t)dt + \Sigma d\mathbf{W}_t, \quad \nu_t = \text{Law}(\mathbf{Z}_t).$$

PROPOSITION 1 Well-posedness

Assume. For any $T > 0$,

$$\nu_0 \in \mathcal{P}_2, \quad \|\mathbf{b}_s(\mathbf{z}, \rho) - \mathbf{b}_s(\mathbf{z}', \rho')\| \leq L_s (\|\mathbf{z} - \mathbf{z}'\| + W_2(\rho, \rho')), \quad s \in \{\text{true}, \text{model}\},$$

and $\nabla \widetilde{K}_h$ is globally Lipschitz; the observation force is uniformly Lipschitz in space.

Then. The assimilated McKean–Vlasov SDE has a **unique strong solution** on $[0, T]$. Consequently, its law ν_t is uniquely determined.

Finite particles converge to the mean-field law

Mean-field particle dynamics

$$d\mathbf{Z}_t = \mathbf{b}_{\text{model}}(\mathbf{Z}_t, \nu_t)dt - \lambda \nabla (\widetilde{K}_h * \nu_t - K_h * \mu_t^{\text{obs}})(\mathbf{Z}_t)dt + \Sigma d\mathbf{W}_t, \quad \nu_t = \text{Law}(\mathbf{Z}_t).$$

Finite- N particle dynamics

$$\nu_t^N = \frac{1}{N} \sum_{j=1}^N \delta_{\mathbf{Z}_t^{j,N}}, \quad d\mathbf{Z}_t^{i,N} = \mathbf{b}_{\text{model}}(\mathbf{Z}_t^{i,N}, \nu_t^N)dt - \lambda \nabla (\widetilde{K}_h * \nu_t^N - K_h * \mu_t^{\text{obs}})(\mathbf{Z}_t^{i,N})dt + \Sigma d\mathbf{W}_t^i.$$

PROPOSITION 2 Propagation of chaos

Assume.

- $\mathbf{b}_s$ is globally Lipschitz in $(\mathbf{x}, W_2)$, $s \in \{\text{true}, \text{model}\}$, and the observation force is uniformly Lipschitz.
- $\nabla \widetilde{K}_h$ is globally Lipschitz and $\|\nabla \widetilde{K}_h\|_{L^\infty} < \infty$.
- The initial ensemble is f_0 chaotic, with $f_0 \in \mathcal{P}_2$.

Then, for every $T > 0$,

$$W_2 \left(\text{Law}(\mathbf{Z}_{[0,T]}^{1,N}), \text{Law}(\mathbf{Z}_{[0,T]}) \right) \longrightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Continuous equations and tracking assumptions

Truth and assimilated density equations

$$\begin{aligned}\partial_t \mu_t &= -\nabla \cdot (\mu_t \mathbf{b}_{\text{true}}(\cdot, \mu_t)) + \nabla \cdot (A \nabla \mu_t), \\ \partial_t \nu_t &= -\nabla \cdot (\nu_t \mathbf{b}_{\text{model}}(\cdot, \nu_t)) + \nabla \cdot (A \nabla \nu_t) + \lambda \nabla \cdot [\nu_t \nabla (K_h * r_t)], \\ r_t &:= K_h * \nu_t - \mu_t^{\text{obs}}, \quad A := \frac{1}{2} \Sigma \Sigma^\top.\end{aligned}$$

THEOREM 3 · ASSUMPTIONS Resolved observations and strong feedback

Assume.

- $\Omega = \mathbb{T}^d$ and $A \succeq \kappa I$, with $\kappa > 0$.
- Exact observations: $\mu_t^{\text{obs}} = K_h * \mu_t$.
- μ_t, ν_t are classical and $\int_{\Omega} (\nu_0 - \mu_0) d\mathbf{x} = 0$.
- $\mathbf{b}_{\text{true}}$ and $\mathbf{b}_{\text{model}}$ are globally Lipschitz in $(\mathbf{x}, W_2)$.
- $\|\mathbf{b}_{\text{model}}\|_{\infty}, \|\nabla \cdot \mathbf{b}_{\text{model}}\|_{\infty} \leq B_{\text{model}}$.
- $\|\mathbf{b}_{\text{model}}(\cdot, \rho) - \mathbf{b}_{\text{model}}(\cdot, \eta)\|_2 \leq L_{\mu} \|\rho - \eta\|_2$.
- $0 < \underline{\nu} \leq \nu_t$ and $\|\mu_t\|_{\infty}, \|\nu_t\|_{\infty} \leq \bar{\rho}$.
- For $e_t := \nu_t - \mu_t$, $\|\nabla e_t - \nabla(\widetilde{K}_h * e_t)\|_2 \leq \delta(h) \|\nabla e_t\|_2$, with $\delta(h) \rightarrow 0$.

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24 / 47

Continuous evolution: L^2 tracking to the model-bias floor

THEOREM 3 · CONCLUSION Tracking error decays to the drift-mismatch floor

Under the assumptions on the previous slide, choose

$$\delta(h) \leq \frac{\underline{\nu}}{2\bar{\rho}}, \quad \lambda > \frac{1}{\underline{\nu}} \left(\frac{2C_{\text{adv}}^2 C_P}{\kappa} - \kappa \right), \quad C_{\text{adv}} := B_{\text{model}} + \bar{\rho} L_{\mu}.$$

$$V(t) := \frac{1}{2} \|\nu_t - \mu_t\|_2^2,$$

$$\Delta := \sup_{s \in [0, T]} \|\mathbf{b}_{\text{model}}(\cdot, \mu_s) - \mathbf{b}_{\text{true}}(\cdot, \mu_s)\|_{\infty},$$

$$\alpha := \frac{\kappa + \lambda \underline{\nu}}{C_P} - \frac{2C_{\text{adv}}^2}{\kappa} > 0.$$

C_P : zero-mean Poincaré constant. From Part I to Part II and back. Linyu, USTC

$$V(t) \leq e^{-\alpha t} V(0) + \frac{\bar{\rho}^2}{\alpha \kappa} \Delta^2, \quad 0 \leq t \leq T$$

$\Delta = 0$. Exponential synchronization.

$\Delta \neq 0$. Squared L^2 error reaches a **bias floor** $O(\Delta^2)$.

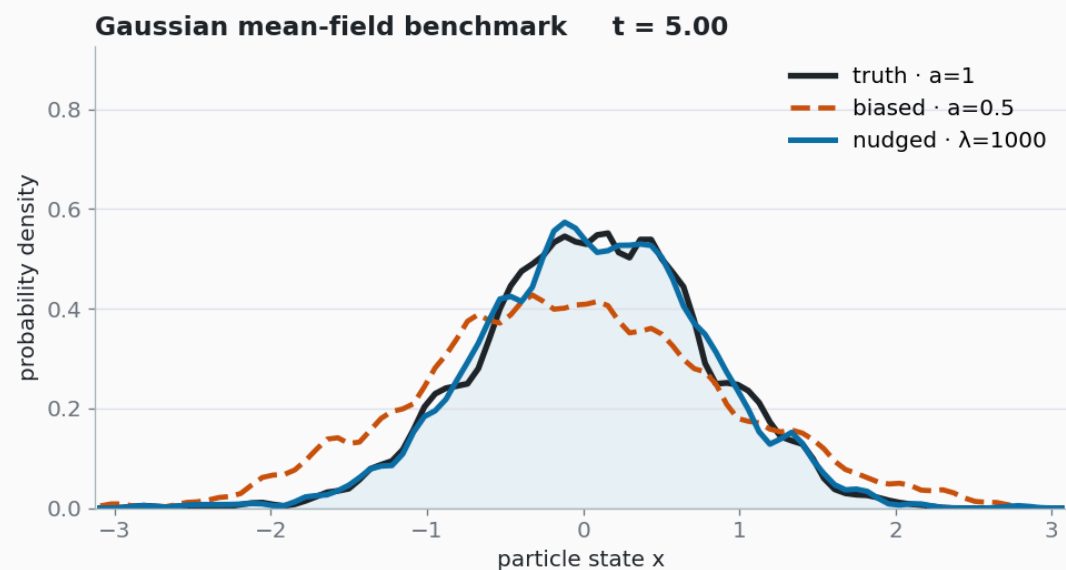
Nudging corrects a misspecified interaction

Design. $dX^i = -a(X^i - m_t)dt + dW^i$, with m_t the empirical mean.

Truth. $a_{\text{true}} = 1$, $X_0^i \sim \mathcal{N}(0, 0.5)$, integrated to $T = 5$.

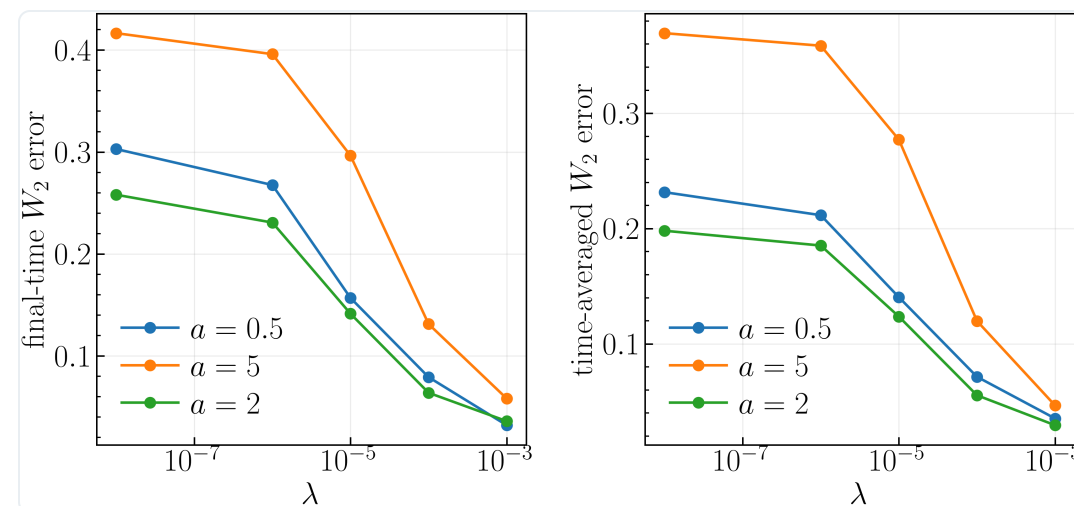
Model. The same SDE with the **wrong coupling** $a \in \{0.5, 2, 5\}$; the animation shows $a = 0.5$.

Observed. The smoothed density $K_h * \mu_t$, with $h = 0.5$.



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Nudging corrects the biased coupling and restores the truth density.



For all three biased couplings, final and time-averaged W_2 errors decrease as λ increases.

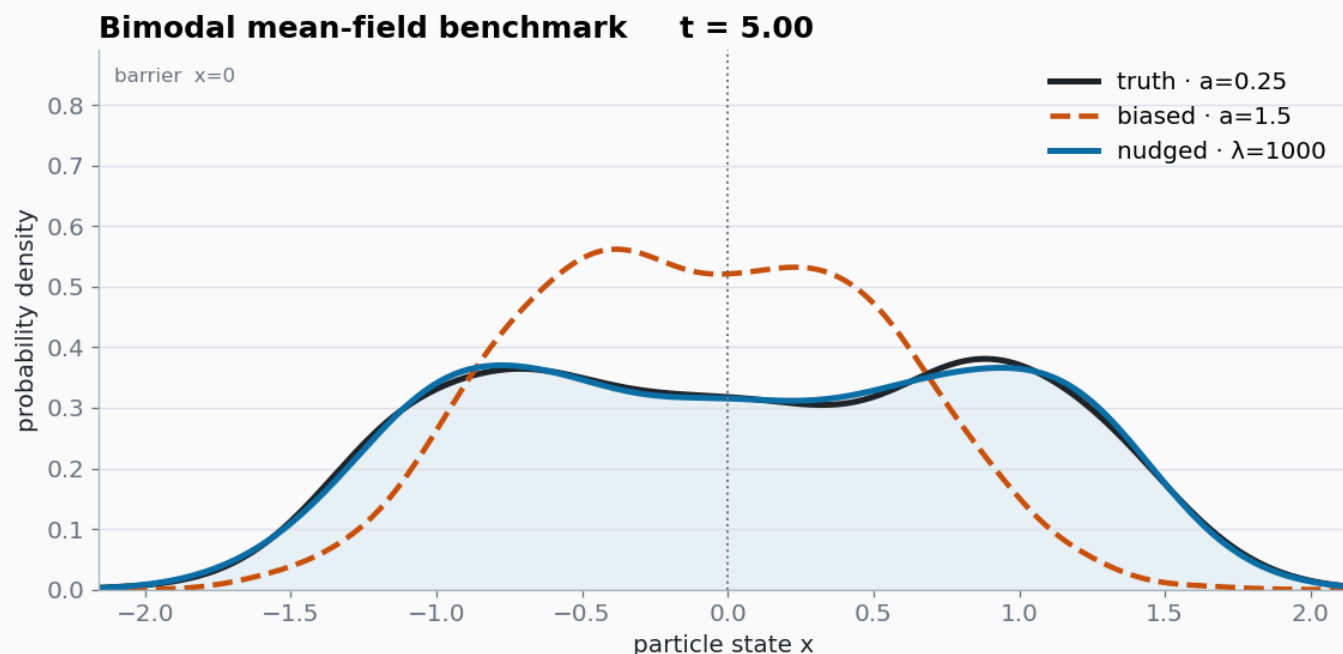
Nudging moves mass across the barrier

Design. $dX^i = -(X_i^3 - X_i)dt - a(X^i - m_t)dt + dW^i$: a double-well system with metastable transitions.

Truth. $a_{\text{true}} = 0.25$, $X_0^i \sim \mathcal{N}(0, 0.5)$, integrated to $T = 5$.

Model. The same dynamics with the wrong coupling; the figure shows $a = 1.5$, which places too much mass near the barrier.

Observed. The smoothed density $K_h * \mu_t$, with $h = 0.5$.



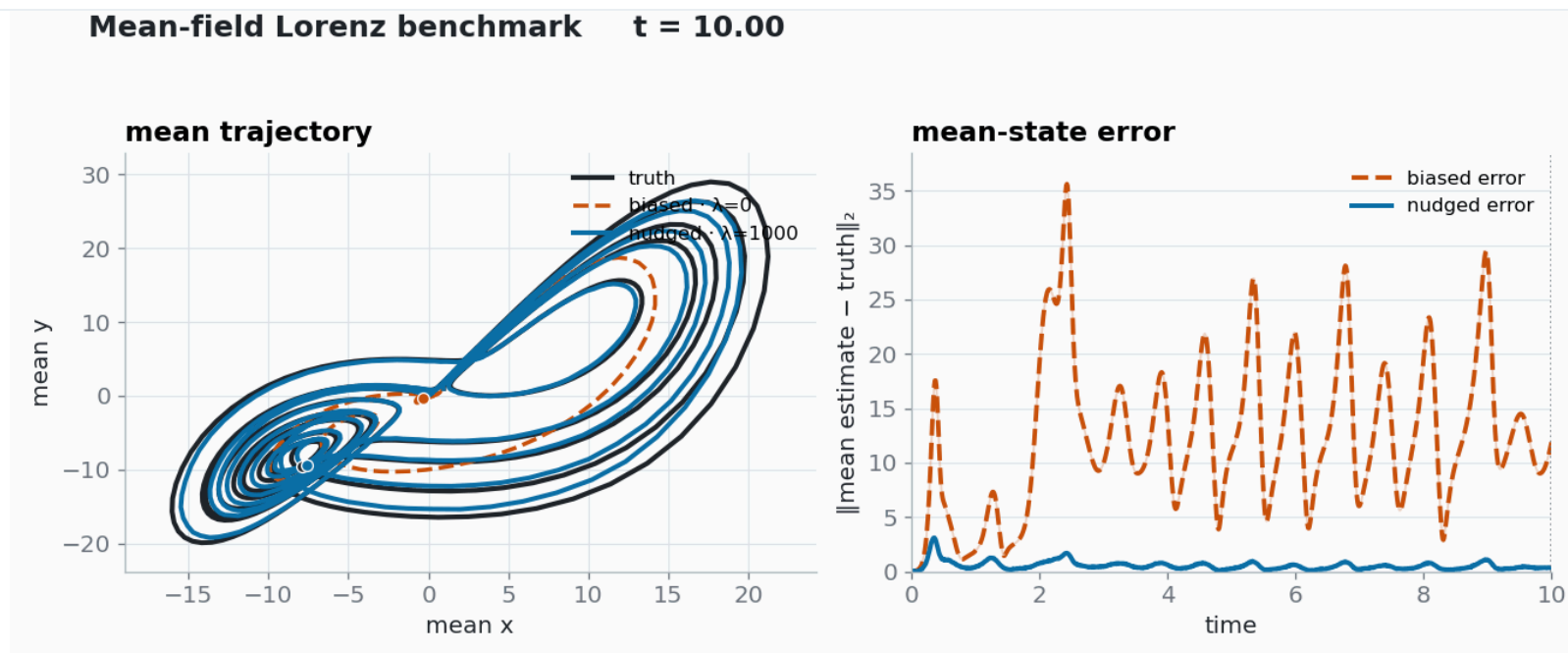
Strong nudging stabilizes a chaotic mean

Design. A stochastic mean-field Lorenz system with $N = 1000$ and coupling through $(m_{t,x}, m_{t,y}, m_{t,z})$.

Truth. Mean-field coupling with $(\sigma_L, \rho, \beta) = (10, 28, 8/3)$ and noise amplitude 1.

Model. N independent Lorenz systems: replace the empirical means by each particle's own coordinates.

Observed. The smoothed three-dimensional density $K_h * \mu_t$, with $h = 0.5$.



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Strong nudging keeps the mean trajectory close to the truth and suppresses forecast error.

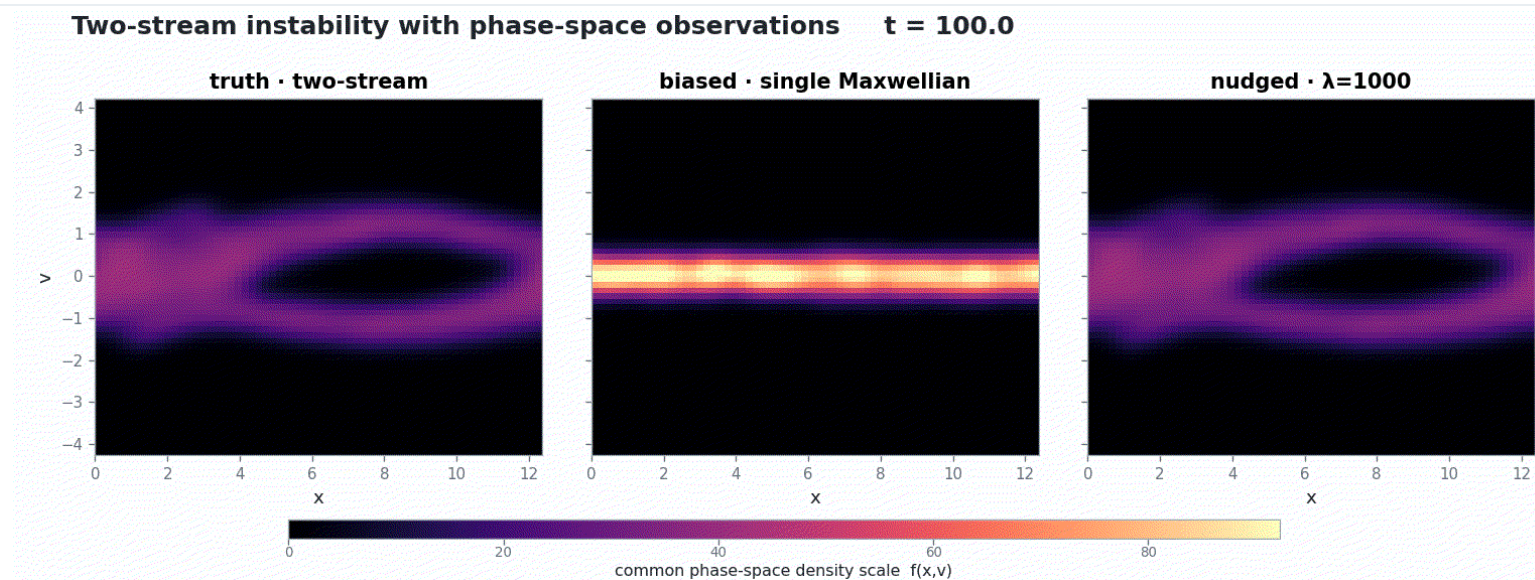
Phase-space data recover the two streams

Design. $\dot{X}^i = V^i$, $\dot{V}^i = E(X^i, t)$, with $E = -\partial_x \phi$ and $-\partial_{xx} \phi = \rho - 1$.

Truth. A perturbed two-stream law with $\sigma = 0.2$, $u_0 = 1$, $\varepsilon = 0.01$, and $\kappa = 0.5$.

Model. The same Vlasov–Poisson dynamics initialized from a **single Maxwellian** — the velocity law is structurally wrong.

Observed. The full smoothed phase-space density $(K_h * f_t)(x, v)$.



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Density nudging reconstructs the two-stream structure missed by the Maxwellian forecast.

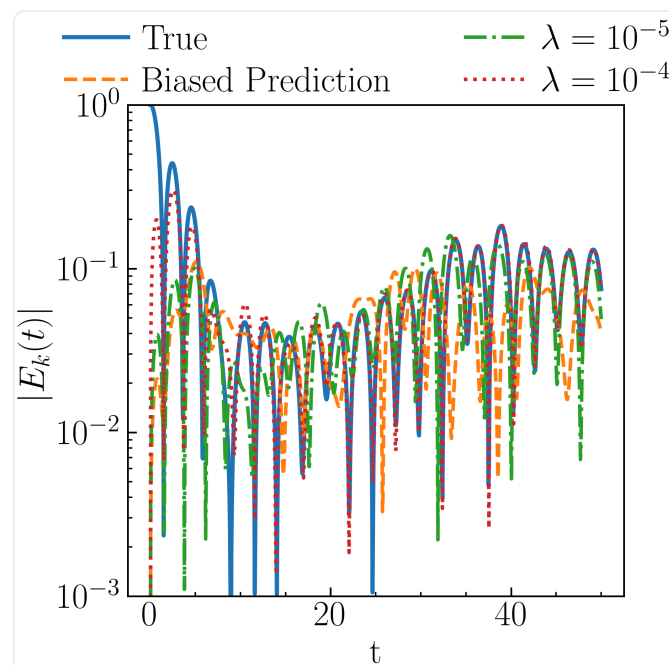
Nudging recovers the Landau damping rate

Design. The same one-dimensional Vlasov–Poisson dynamics, monitored through the fundamental electric-field mode $|\hat{E}(\kappa, t)|$.

Truth. $f_0 \propto f_{M,1}(v)(1 + \varepsilon \cos \kappa x)$, $\varepsilon = 0.5$, $\kappa = \frac{1}{2}$, $x \in [0, 4\pi]$.

Model. The same dynamics initialized from the unperturbed Maxwellian, $\varepsilon = 0$ — no initial spatial modulation.

Observed. The full smoothed phase-space density $(K_h * f_t)(x, v)$.

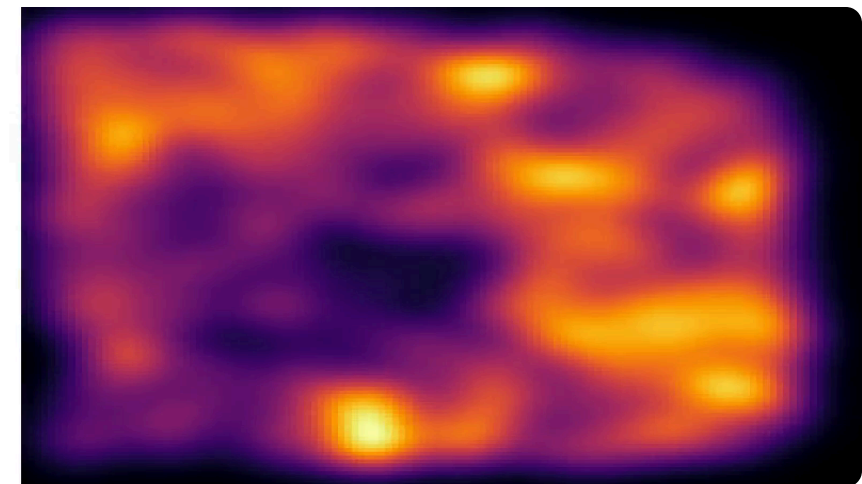
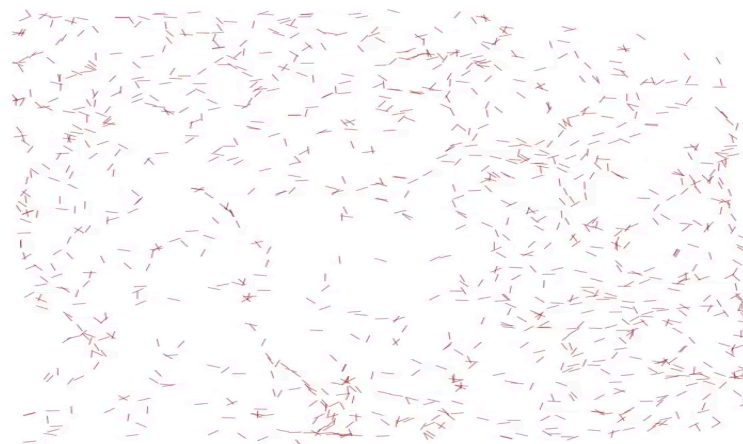


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Reference / biased / assimilated: the nudged runs recover the damping profile from density-level data.

30 / 47

Density feedback stabilizes a learned fish model



Multiscale nudging keeps the learned fish forecast aligned with the observed density.

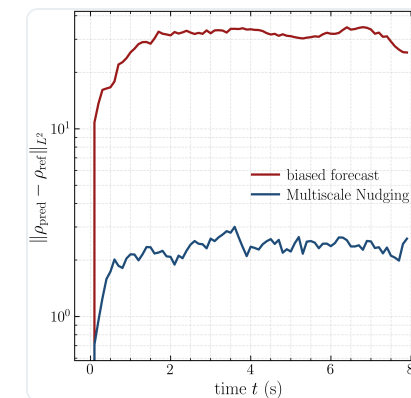
Design. Start all three runs from the same frame; integrate with $\Delta t = 0.025$ s, $L_{\text{nud}} = 100$, and $\lambda = 1$.

Truth. TRex tracks for $N \approx 1126$ fish, with occlusions, missing detections, and inconsistent identities.

Model. The learned MVNN drift from Part I, rolled out without observation feedback.

Observed. Gaussian KDE of the unmasked detections on a 125×125 grid, with $h = 2$ px.

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Density L^2 error: biased saturates near 30 within 1 s; nudged stays near 2 – 10× lower.

Moment nudging for Vlasov–Poisson

Correct particles from $(\rho, \mathbf{u}, T)$ without reconstructing the full phase-space law.

Moments determine a family, not a distribution

$$\begin{aligned}\partial_t f + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \mathbf{E}[f] \cdot \nabla_{\mathbf{v}} f &= Q[f], \\ \mathbf{E}[f] &= -\nabla_{\mathbf{x}} \phi, \quad -\Delta_{\mathbf{x}} \phi = \rho[f] - 1.\end{aligned}$$

PIC: $f^{N_p} = \sum_i w_i \delta_{(\mathbf{x}^i, \mathbf{v}^i)}$ — the particles are quadrature points of one law.

Observed moment map — linear in f

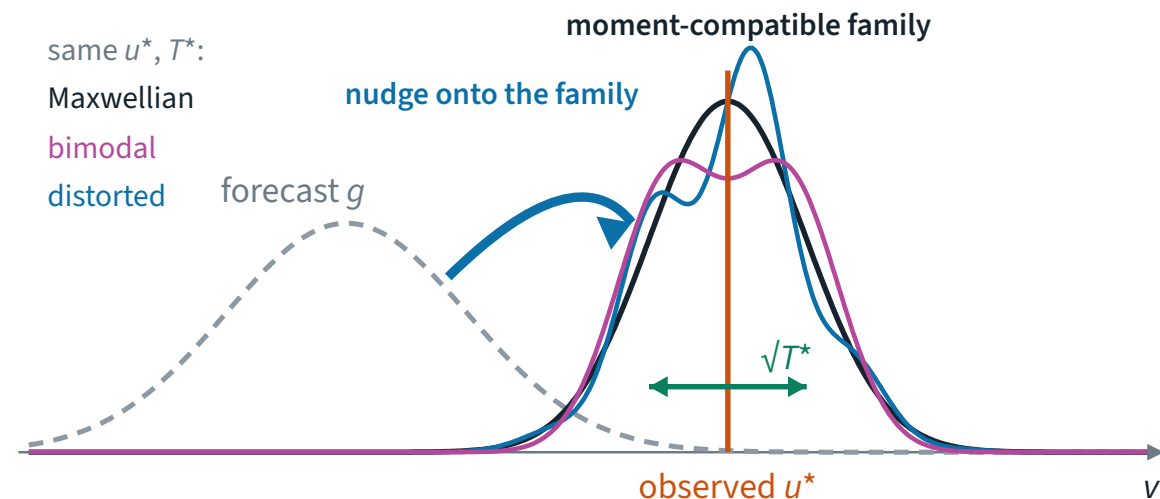
$$M_h[f](\mathbf{x}) := (\rho_h, \mathbf{j}_h, \mathcal{K}_h)[f](\mathbf{x}) = \int K_h(\mathbf{x} - \mathbf{y}) (1, \mathbf{v}, \tfrac{1}{2}|\mathbf{v}|^2) f(\mathbf{y}, \mathbf{v}) \, d\mathbf{y} \, d\mathbf{v}.$$

$$\mathbf{r}[f] := M_h[f] - M^{\text{obs}} = (r_0, \mathbf{r}_1, r_2),$$

$$J_M[f] := \frac{\gamma_1}{2} \|r_0\|_{L^2}^2 + \frac{\gamma_2}{2} \|\mathbf{r}_1\|_{L^2}^2 + \frac{\gamma_3}{2} \|r_2\|_{L^2}^2.$$

Moment-compatible family $\mathcal{S}_h = \{f : J_M[f] = 0\} \ni f^\dagger$. The data fix this family, not one distribution; feedback must therefore **vanish on all of $\mathcal{S}_h$** .

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At one $\mathbf{x}$: the forecast g has the wrong moments; feedback must move it onto the family sharing $\mathbf{u}^*, \mathbf{T}^*$. Collisions and the dynamics then move the unobserved shape along the family.

Method A: weight the transport geometry

Moment residuals define a phase-space potential

$$\Phi_M[f] := \frac{\delta J_M}{\delta f} = \gamma_1 q_0(\mathbf{x}) + \gamma_2 \mathbf{v} \cdot \mathbf{q}_1(\mathbf{x}) + \frac{\gamma_3}{2} |\mathbf{v}|^2 q_2(\mathbf{x}), \quad (q_0, \mathbf{q}_1, q_2) = K_h * (r_0, \mathbf{r}_1, r_2).$$

Define the metric tensor G_t

$$G_t(\mathbf{x}, \mathbf{v}) = \text{diag}(a_t(\mathbf{x}, \mathbf{v})I_d, I_d), \quad a_t = 1 + \frac{|\mathbf{v} - \mathbf{u}^{\text{obs}}(\mathbf{x}, t)|^2}{V_*^2}.$$

V_* is a reference thermal speed. The factor a_t increases only the cost of spatial motion.

Descend J_M in that metric

$$\begin{aligned} \mathbf{b}_A &= -G_t^{-1} \nabla_{\mathbf{x}, \mathbf{v}} \Phi_M \\ &= \left(-\frac{\nabla_{\mathbf{x}} \Phi_M}{1 + |\mathbf{v} - \mathbf{u}^{\text{obs}}|^2 / V_*^2}, -\nabla_{\mathbf{v}} \Phi_M \right). \end{aligned}$$

Large peculiar velocity damps the position channel; the velocity channel is unchanged.

Why introduce G_t ? Plain W_2 leaves $\frac{\gamma_3}{2} |\mathbf{v}|^2 \nabla_{\mathbf{x}} q_2$ in the position drift. Dividing that channel by a_t controls the quadratic growth, while $\frac{d}{dt} J_M[f_t] \leq 0$.

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Method B: split residuals by direction

Keep the same quadratic mismatch, but split it

$$J_M = \underbrace{\frac{\gamma_1}{2} \|r_0\|_{L^2}^2}_{J_M^\rho} + \underbrace{\frac{\gamma_2}{2} \|\mathbf{r}_1\|_{L^2}^2 + \frac{\gamma_3}{2} \|r_2\|_{L^2}^2}_{J_M^{\mathbf{j}, \kappa}}, \quad (q_0, \mathbf{q}_1, q_2) = K_h * (r_0, \mathbf{r}_1, r_2).$$

Density residual $\rightarrow$ position

$$\Phi_M^\rho = \gamma_1 q_0(\mathbf{x}), \quad \mathbf{b}_{B, \mathbf{x}} = -\nabla_{\mathbf{x}} \Phi_M^\rho = -\gamma_1 \nabla_{\mathbf{x}} q_0.$$

Particles move spatially to correct the observed density ρ^{obs} .

Momentum + energy $\rightarrow$ velocity

$$\Phi_M^{\mathbf{j}, \kappa} = \gamma_2 \mathbf{v} \cdot \mathbf{q}_1 + \frac{\gamma_3}{2} |\mathbf{v}|^2 q_2,$$

$$\mathbf{b}_{B, \mathbf{v}} = -\nabla_{\mathbf{v}} \Phi_M^{\mathbf{j}, \kappa} = -\gamma_2 \mathbf{q}_1 - \gamma_3 \mathbf{v} q_2.$$

Velocity translations correct momentum; velocity scaling corrects kinetic energy.

No metric tensor is needed. Method B uses plain W_2 , but the energy residual never enters the position update; therefore the problematic $|\mathbf{v}|^2 \nabla_{\mathbf{x}} q_2$ term is absent.

Method C: penalize moments, not shape

Why full relative entropy is too strong

Let $\pi^{\text{obs}} = \mathcal{M}_{\rho^{\text{obs}}, \mathbf{u}^{\text{obs}}, \Theta^{\text{obs}}}$ and let $\mathcal{M}[f]$ be the Maxwellian with the same $(\rho_h, \mathbf{u}_h, \Theta_h)$ as f .

$$\text{KL}(p||q) := \int \left(p \log \frac{p}{q} - p + q \right) \geq 0.$$

$$\text{KL}(f||\pi^{\text{obs}}) = \underbrace{\text{KL}(f||\mathcal{M}[f])}_{\text{unobserved shape}} + \underbrace{\text{KL}(\mathcal{M}[f]||\pi^{\text{obs}})}_{\text{moment mismatch}}.$$

Using the full KL would keep pushing every non-Maxwellian f toward π^{obs} , even after its observed moments are correct.

Keep only the moment term

$$J_{\text{ent}}[f] := \text{KL}(\mathcal{M}_{h,\varepsilon}[f] || \pi_{\varepsilon}^{\text{obs}}), \quad \Theta_h = T_h + \varepsilon, \quad \Theta^{\text{obs}} = T^{\text{obs}} + \varepsilon.$$

$J_{\text{ent}} = 0$ exactly when $(\rho_h, \mathbf{u}_h, T_h) = (\rho^{\text{obs}}, \mathbf{u}^{\text{obs}}, T^{\text{obs}})$: the whole moment-compatible family remains a zero set.

Weighted-Wasserstein feedback

$$\mathbf{b}_{C,\mathbf{x}} = -\gamma \frac{\nabla_{\mathbf{x}} \Psi_{\varepsilon}}{1 + |\mathbf{v} - \mathbf{u}^{\text{obs}}|^2 / V_*^2}, \quad \Psi_{\varepsilon} = \frac{\delta J_{\text{ent}}}{\delta f},$$

$$\mathbf{b}_{C,\mathbf{v}} = -\gamma K_h * \left[\frac{\mathbf{v} - \mathbf{u}^{\text{obs}}}{\Theta^{\text{obs}}} - \frac{\mathbf{v} - \mathbf{u}_h}{\Theta_h} \right].$$

Why shape is preserved. $\mathbf{b}_{C,\mathbf{v}}$ is affine in $\mathbf{v}$: it translates and rescales velocity to correct $\mathbf{u}$ and T , while leaving the unresolved non-Maxwellian shape term $\text{KL}(f||\mathcal{M}[f])$ unchanged.

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All three schemes are well posed

A • Lyapunov structure

$$\frac{d}{dt} J_M[f_t] = -\|\text{grad}_G J_M\|^2 \leq 0.$$

One mismatch, one weighted gradient flow.

B • No quadratic position drift

Direction splitting removes the problematic

$$|\mathbf{v}|^2 \nabla_{\mathbf{x}} (K_h * r_2)$$

from the position update.

C • Closed moment balances

$$\dot{\mathbf{u}} = -\frac{\gamma}{T_{\text{obs}}} (\bar{\mathbf{u}} - \mathbf{u}^{\text{obs}}).$$

Affine velocity maps are shape preserving.

PROPOSITION Global well-posedness for A, B and C

With Lenard–Bernstein collisions, smooth positive kernels and observations, and finite initial second moment, the N_p -particle SDE has a **unique global strong solution** for every method $a \in \{A, B, C\}$ and every feedback strength.

Same temperature, different current and shape

Undriven Vlasov–Poisson–Dougherty

$$\partial_t h + v \partial_x h + E[h] \partial_v h = Q_D[h].$$

$$-\partial_{xx} \phi = \rho[h] - 1, \quad E = -\partial_x \phi.$$

$$Q_D[h] = \nu \partial_v ((v - u[h])h + T[h] \partial_v h).$$

$$\rho[h] = \int h \, dv, \quad u[h] = \rho[h]^{-1} \int v h \, dv, \quad T[h] = \rho[h]^{-1} \int (v - u[h])^2 h \, dv.$$

Truth · Maxwellian current wave

$$f_0^\dagger(x, v) = \mathcal{M}_{1, u_0^\dagger(x), T_0}(v),$$

$$u_0^\dagger(x) = 0.25 + 0.8 \sin(x/2), \quad T_0 = 0.5.$$

$$\mathcal{M}_{1, u, T}(v) = \frac{1}{\sqrt{2\pi T}} \exp\left[-\frac{(v - u)^2}{2T}\right].$$

Unit density · single Gaussian velocity shape.

Prior · shifted bimodal current wave

$$g_0(x, v) = \frac{1}{2} \mathcal{M}_{1, u_0(x) - a, \theta}(v) + \frac{1}{2} \mathcal{M}_{1, u_0(x) + a, \theta}(v),$$

$$u_0(x) = -0.25 - 0.3 \sin(x/2 + \pi/3),$$

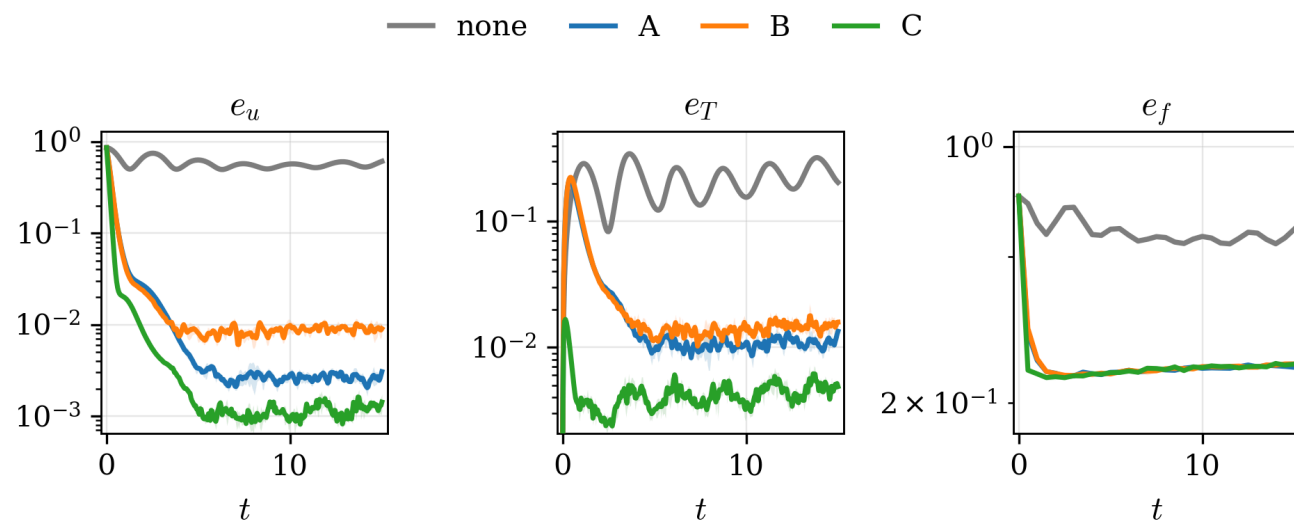
$$a = 0.55, \quad \theta = T_0 - a^2 = 0.1975.$$

Same $\rho_0 = 1$ and $T_0 = 0.5$; wrong mean flow, amplitude, phase, and velocity-space shape.

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Observed $(\rho, u, T)(x, t)$, every step Grid $L = 4\pi$, $N_p = 10^5$, $N_x = 128$ Run $\Delta t = 0.002$, $T_{\text{fin}} = 15$, $\nu = 0.5$ Feedback $\gamma_{1,2,3} = \gamma = 3$, $V_* = 1$ (A,C)

Moment errors collapse; shape relaxes more slowly



Methods A/B/C rapidly reduce the moment errors relative to no assimilation.

Mean error / no assimilation

$R_*^S(10, 15)$, averaged over $t \in [10, 15]$ and three seeds.

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METHOD	R_ρ	R_u	R_T	R_f
A	0.015	0.005	0.045	0.444
B	0.008	0.016	0.061	0.447
C	0.012	0.002	0.019	0.446

39 / 47

The prior reverses, shifts, and weakens the vortex

Undriven 2D2V Vlasov–Poisson–Dougherty

$$\partial_t h + \mathbf{v} \cdot \nabla_{\mathbf{x}} h + \mathbf{E}[h] \cdot \nabla_{\mathbf{v}} h = Q_D[h].$$

$$-\Delta_{\mathbf{x}} \phi = \rho[h] - 1, \quad \mathbf{E} = -\nabla_{\mathbf{x}} \phi.$$

$$Q_D[h] = \nu \nabla_{\mathbf{v}} \cdot ((\mathbf{v} - \mathbf{u}[h])h + T[h] \nabla_{\mathbf{v}} h).$$

$$\rho[h] = \int h \, d\mathbf{v}, \quad \mathbf{u}[h] = \rho[h]^{-1} \int \mathbf{v} h \, d\mathbf{v}, \quad T[h] = (2\rho[h])^{-1} \int |\mathbf{v} - \mathbf{u}[h]|^2 h \, d\mathbf{v}.$$

Truth · Taylor–Green Maxwellian

$$f_0^\dagger(\mathbf{x}, \mathbf{v}) = \mathcal{M}_{1, \mathbf{u}_0^\dagger(\mathbf{x}), 0.5}(\mathbf{v}),$$

$$u_{0,x}^\dagger(x, y) = 0.25 + \sin x \cos y,$$

$$u_{0,y}^\dagger(x, y) = -\cos x \sin y.$$

Mean flow $(0.25, 0)$ · unit vortex amplitude · isotropic covariance $0.5I_2$.

Prior · anisotropic Gaussian

$$g_0(\mathbf{x}, \mathbf{v}) = \frac{\exp\left[-\frac{1}{2}(\mathbf{v} - \mathbf{u}_0)^\top \Sigma_0^{-1}(\mathbf{v} - \mathbf{u}_0)\right]}{2\pi\sqrt{\det \Sigma_0}},$$

$$\Sigma_0 = \text{diag}(0.8, 0.2),$$

$$u_{0,x} = -0.25 - 0.45 \sin(x + \pi/3) \cos(y + \pi/4),$$

$$u_{0,y} = 0.15 + 0.45 \cos(x + \pi/3) \sin(y + \pi/4).$$

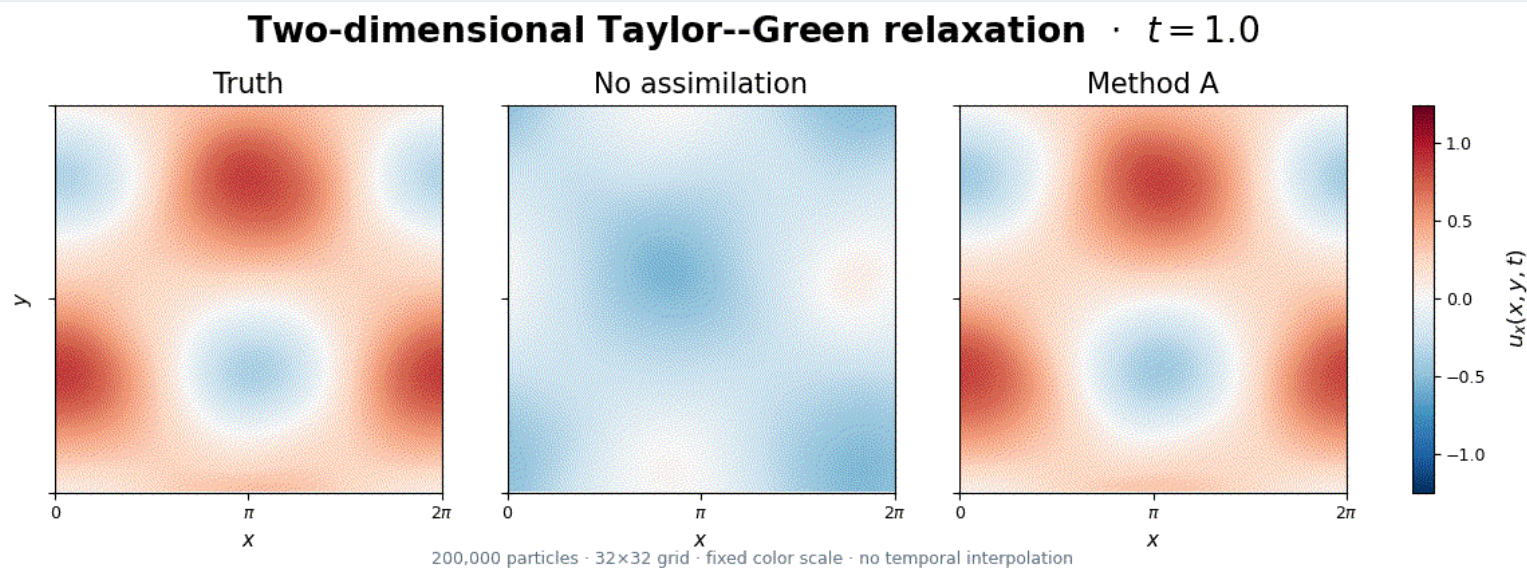
Wrong mean flow, phase, rotation, amplitude, and velocity anisotropy; but $\frac{1}{2} \text{tr} \Sigma_0 = 0.5$, so the scalar temperature is initially correct.

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Observed $(\rho, \mathbf{u}, T)(x, y, t)$, every step Grid $N_p = 2 \times 10^5, 32^2$ Run $\Delta t = 0.002, T_{\text{fin}} = 12, \nu = 0.5$ Domain $[0, 2\pi)^2$ Feedback $\gamma_{1,2,3} = \gamma = 3, V_* = 1$ (A,C)

40 / 47

Method A follows the vortex through relaxation



Method A recovers the true vortex cells and follows their collisional relaxation.

Late-time benchmark

$R_*^S(8, 12)$: mean error / no assimilation over $t \in [8, 12]$
and three seeds · Particles to Measures and Back · Liyao Lyu, USTC

METHOD	R_ρ	R_u	R_T
A	0.293	0.019	0.039
B	0.266	0.027	0.044
C	0.393	0.020	0.032

The two-stream truth is far from equilibrium

Self-consistent 1D1V Vlasov–Poisson–BGK · no external driver

$$\partial_t h + v \partial_x h + E[h] \partial_v h = Q_{\text{BGK}}[h].$$

$$-\partial_{xx} \phi = \rho[h] - 1, \quad E = -\partial_x \phi.$$

$$Q_{\text{BGK}}[h] = \nu(\mathcal{M}[h] - h), \quad \mathcal{M}[h] = \frac{\rho[h]}{\sqrt{2\pi T[h]}} \exp\left[-\frac{(v - u[h])^2}{2T[h]}\right].$$

$$\rho[h] = \int h \, dv, \quad u[h] = \rho[h]^{-1} \int v h \, dv, \quad T[h] = \rho[h]^{-1} \int (v - u[h])^2 h \, dv.$$

Truth · unstable two-stream state

$$f_0^\dagger(x, v) = \frac{1}{\sqrt{2\pi}} v^2 e^{-v^2/2} (1 + 0.05 \cos(x/2)), \quad \rho_0^\dagger(x) = 1 + 0.05 \cos(x/2), \quad u_0^\dagger = 0, \quad T_0^\dagger = 3.$$

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Observed $(\rho, u, T)(x, t)$, every step Domain $L = 4\pi$, $\alpha = 0.05$, $k_0 = 0.5$

Grid $N_p = 5 \times 10^5$, $N_x = 128$

Run $\Delta t = 0.05$, $T_{\text{fin}} = 30$

Truth BGK $\nu = 0.05$

Same wrong (u, T) , different velocity shapes

Truth: $(u_0^\dagger, T_0^\dagger) = (0, 3)$. Both priors: $(u_0, T_0) = (0.3, 2)$.

Prior 1 · Maxwellian

$$g_0^{\text{M}}(x, v) = \mathcal{M}_{1,0.3,2}(v).$$

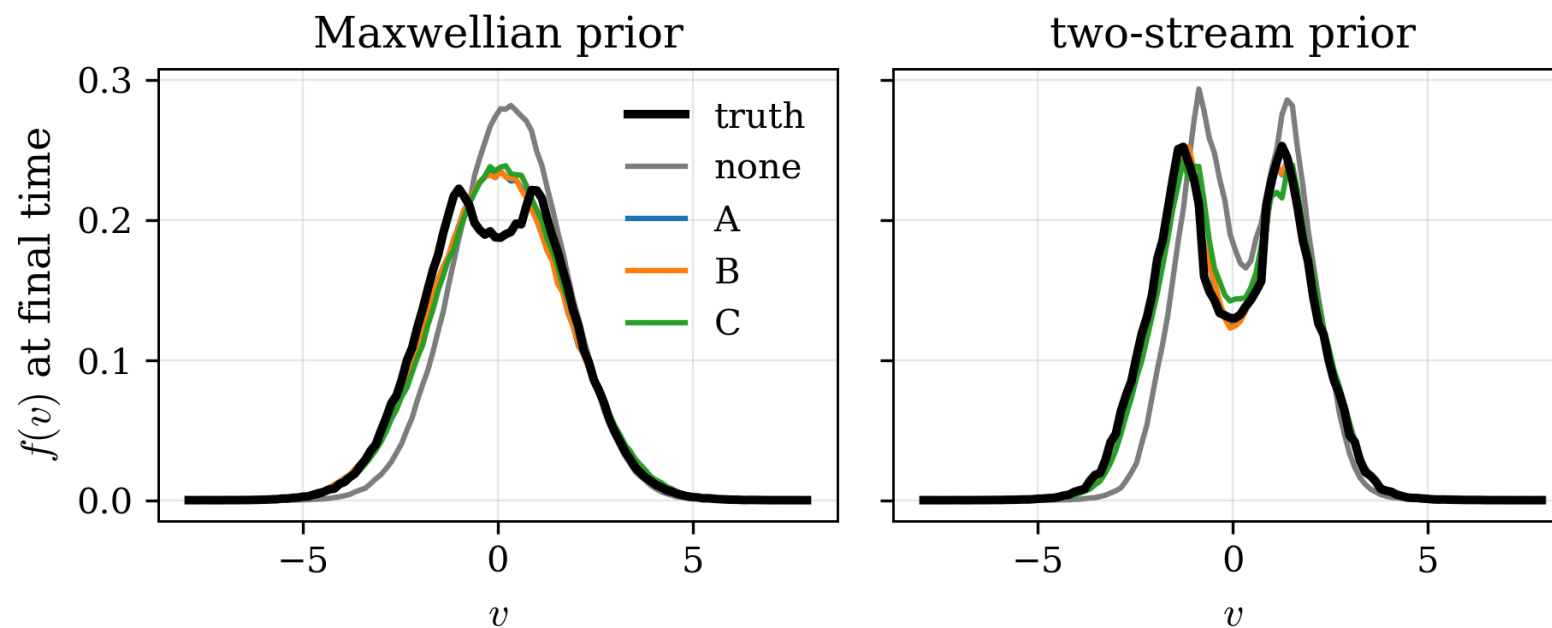
Uniform in x · single-Gaussian velocity shape · no seeded density modulation.

Prior 2 · affine two-stream

$$g_0^{2s} = (\text{Id}_x, A)_\# f_0^\dagger, \quad A(v) = 0.3 + \sqrt{2/3} v.$$

The seeded density modulation and two-stream velocity shape are retained.

Prior 1 corrects the moments—but stays single-peaked



A Maxwellian prior stays single-peaked; a two-stream prior preserves the missing bimodality.

Maxwellian prior

A/B/C improve (u, T) , but the marginal remains single-peaked and misses the bimodal truth.

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Two-stream prior

A/B/C preserve the two peaks; the missing velocity-space shape was already present in the prior.

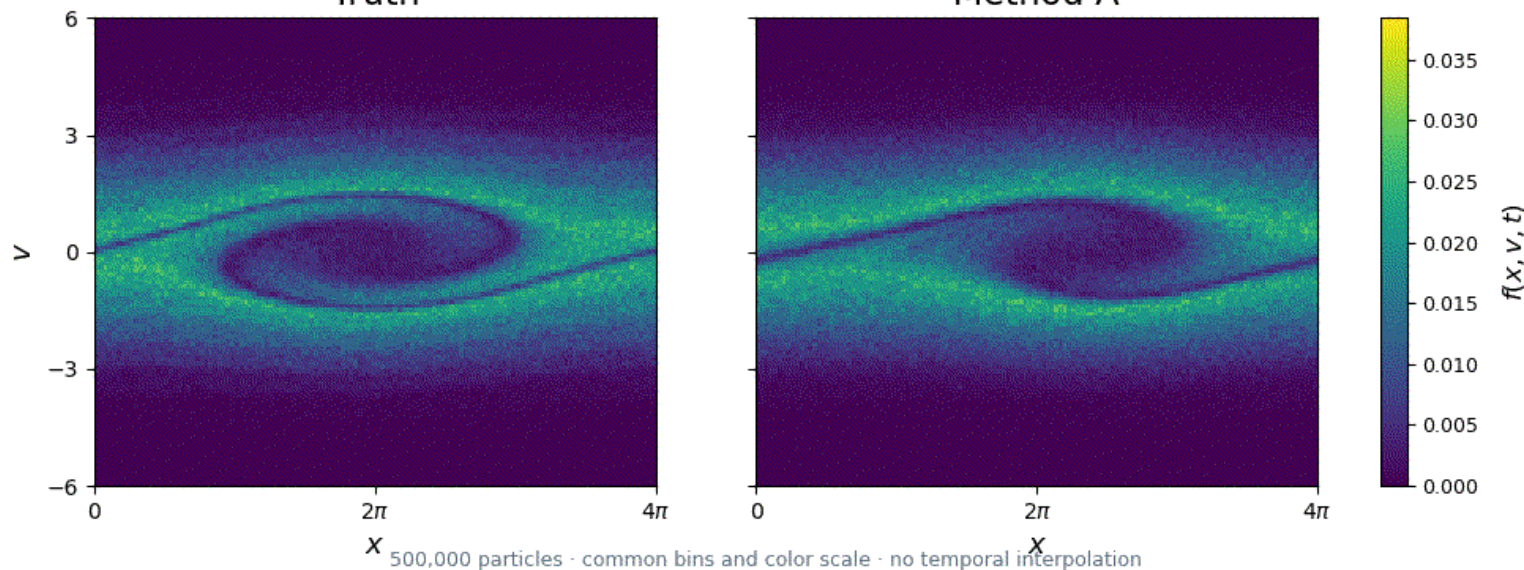
44 / 47

With the right shape, Method A tracks the vortex

Two-stream phase space · $t = 20$

Truth

Method A



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With the correct prior shape, Method A tracks the evolving phase-space vortex.

OBSERVED

The moments synchronize.

The feedback corrects the wrong $(u, T) = (0.3, 2)$ toward the truth $(0, 3)$.

NOT OBSERVED

The shape must come from the prior.

A two-stream prior develops the vortex; a Maxwellian prior cannot become bimodal from moment data alone.

Boundary. With the right shape in the prior, Method A tracks the full distribution at $R_f = 0.48$; the feedback does not invent that shape. 45 / 47

One principle, three instances

	I. MVNN — LEARN	II. MULTISCALE NUDGING — CORRECT	III. MOMENT NUDGING — CORRECT, PLASMA
measure-level object	drift $\mathbf{b}(\mathbf{x}, \mu)$ on $\mathbb{R}^d \times \mathcal{P}_2$	misfit $J = \frac{1}{2} \int K_h * \nu - \mu^{\text{obs}} ^2$	moment mismatch J_M / moment relative entropy; zero set = $\mathcal{S}_h$
particle realization	$\varphi_{\text{int}}(\mathbf{x}, \frac{1}{N} \sum_j \varphi_{\text{emb}}(\mathbf{X}^j)), O(N)$	$-\lambda \nabla(\widetilde{K}_h * \nu^N - K_h * \mu^{\text{obs}})(\mathbf{Z}^i)$	$-G_t^{-1} \nabla_{\mathbf{x}, \mathbf{v}} \Phi_M$ (weighted W_2); split; entropic
guarantees	well-posed, propagation of chaos, universal approximation, rates under order-parameter structure	well-posed, propagation of chaos, L^2 decay to a bias floor $\propto \Delta^2$	two-Maxwellian obstruction, Lyapunov, global moment balances, global well-posedness (Lenard–Bernstein)
evidence	Motsch–Tadmor, order-parameter drifts, 2D swarms on unseen geometries, multi-group, second order	Gaussian / bimodal / Lorenz / Vlasov–Poisson / a real fish school (10× lower density error)	1D1V & 2D2V: $\mathbf{u}, \mathbf{T}$ errors down 10–100×; two-stream delimits what moments can recover

Outlook. Noisy, sparse and intermittent observations; synchronization and mean-field limits of the nudged PIC scheme; magnetic fields and non-periodic boundaries; learning beyond mean field (two-particle correlations, kinetic closures); **learning and assimilating online** — update $\mathbf{b}_\theta$ while nudging.

Questions?

THIS TALK

- **L. Lyu, X. Yu, H. Schaeffer.** *MVNN: A measure-valued neural network for learning McKean–Vlasov dynamics from particle data.* arXiv:2604.00333 (2026).
- **L. Lyu, X. Yu, H. Schaeffer.** *Multiscale Nudging: From macroscopic observations to microscopic dynamics.* arXiv:2606.06809 (2026). Code: github.com/Lyuliyao/multiscale-nudging
- **L. Lyu, X. Yu, D. Schneideringer, H. Schaeffer.** *Wasserstein moment nudging for Vlasov–Poisson data assimilation.* Preprint (2026).

Computations used Anvil (Purdue) through ACCESS allocation MTH260007 and the NVIDIA Academic Grant Program.

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- Azouani, Olson, Titi (2014); Bessaih, Olson, Titi (2015), continuous data assimilation · Jordan, Kinderlehrer, Otto (1998); Ambrosio, Gigli, Savaré (2005), Wasserstein gradient flows · Benamou & Brenier (2000).
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